



Year 11 Mathematics Extension 1
Assessment Task 1
Term 2, Week 5, 2021

Name: _____

Teacher: _____

Friday 21st May, 2021

Setter: WIL

General Instructions

- **Working time – 45 minutes.**
- Answer all questions on the lined paper provided using black or blue pen.
- NESA approved calculators may be used.
- Show all relevant mathematical reasoning and/or calculations.
- **Total marks – 32.**
- Attempt Questions 1 – 5.

Question 1 (6 marks)

Marks

For **Multiple Choice** questions, write down the correct option on your writing paper.

- (a) Which one of the following transformations is **not** required to obtain $g(x) = -1 + \frac{1}{2}f(3 - 2x)$ from $f(x)$? **1**

- A. A reflection in the y -axis
- B. A reflection in the x -axis
- C. A translation parallel to the x -axis
- D. A compression in the y direction.

- (b) Suppose that the equation $x^2 + 5x - 1 = 0$ has roots α and β . Without solving the equation, find the values of:

- (i) $\alpha + \beta$ **1**
- (ii) $\alpha\beta$ **1**
- (iii) $(\alpha - \beta)^2$ **2**
- (iv) $|\alpha - \beta|$ **1**

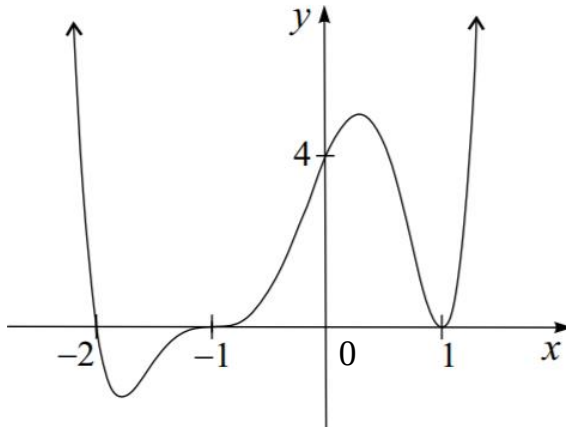
End of Question 1

Question 2 (6 marks) (Start a new page)**Marks**

For **Multiple Choice** questions, write down the correct option on your writing paper.

- (a) The diagram below shows the graph of $y = a(x+b)(x+c)^2(x+d)^3$.

What are the possible values of a, b, c and d ?

1

- A. $a = 1, b = 1, c = -2, d = -1$
- B. $a = 1, b = 2, c = -1, d = 1$
- C. $a = 2, b = 2, c = -1, d = 1$
- D. $a = 2, b = 2, c = 1, d = -1$.

- (b) Solve $\sin 2\theta = -\frac{1}{2}\sqrt{3}$, for $-90^\circ \leq \theta \leq 90^\circ$.

2

- (c) (i) Without the use of calculus, sketch (using quarter of a page) the polynomial $y = x(x^2 - 1)$ showing all the intercepts with the axes.

1

- (ii) Hence, or otherwise, solve the inequation $\frac{x(x+1)}{(x-1)} \geq 0$.

2**End of Question 2**

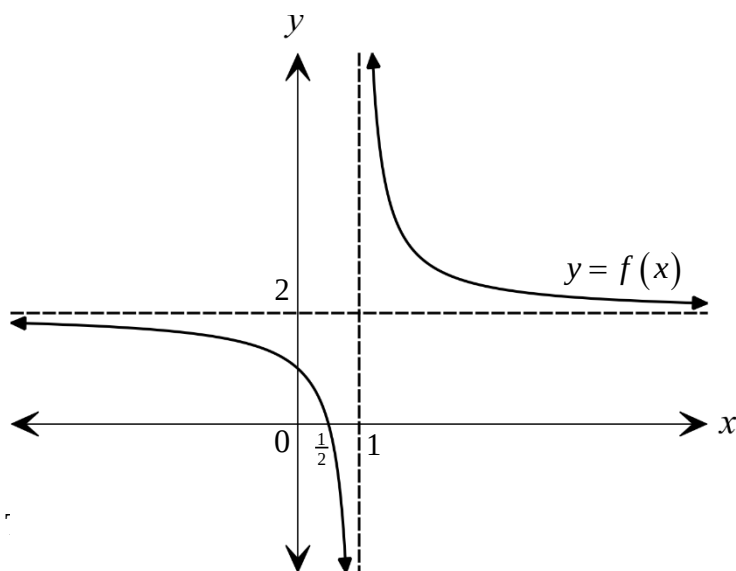
For this **Multiple Choice** question, write down the correct option on the yellow sheet for Question 3.

- (a) A curve is defined parametrically by the equations $x = \sin \theta$, $y = 2 \cos \theta$.
What is the Cartesian equation of the curve?

1

- A. $4x^2 + y^2 = 4$
B. $4x^2 + y^2 = 1$
C. $x^2 + y^2 = 1$
D. $x^2 + y^2 = 4$.

(b)



The diagram above shows the graph of $y = f(x)$, which has a horizontal asymptote of $y = 2$ and a vertical asymptote of $x = 1$. Use the separate yellow sheet for Question 3 to sketch the following graphs, clearly showing any intercepts with axes and asymptotes. Please go over $f(x)$ on the template with your pen where it is part of your solution.

- (i) $y = |f(x)|$ 1
(ii) $y = f(|x|)$ 1
(iii) $|y| = f(x)$ 1
(iv) $y = f(x) + g(x)$, where $g(x) = x$. 2

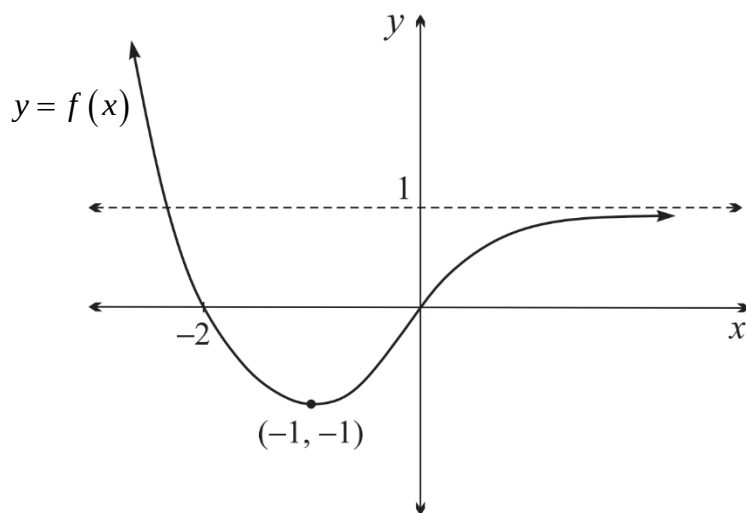
End of Question 3

For this **Multiple Choice** question, write down the correct option on the green sheet for Question 4.

- (a) A function is defined by $f(x) = -\sqrt{4-x^2}$ for $0 \leq x \leq 2$. Which of the following correctly represents the inverse function of $f(x)$? **1**

- A. $f^{-1}(x) = -\sqrt{4-x^2}$ for $0 \leq x \leq 2$
- B. $f^{-1}(x) = \sqrt{4-x^2}$ for $0 \leq x \leq 2$
- C. $f^{-1}(x) = -\sqrt{4-x^2}$ for $-2 \leq x \leq 0$
- D. $f^{-1}(x) = \sqrt{4-x^2}$ for $-2 \leq x \leq 0$.

(b)



The diagram above shows the graph of $y = f(x)$, which has a horizontal asymptote of $y = 1$.

Use the separate green sheet for Question 4 to sketch the following graphs, clearly showing any intercepts with axes and asymptotes. Please go over $f(x)$ on the template with your pen, where it is part of your solution.

(i) $y = \frac{1}{f(x)}$ **1**

(ii) $y^2 = f(x)$ **2**

- (c) (i) A polynomial $Q(x)$ is defined by $Q(x) = x^3 - b^3$, where b is a constant. Divide $Q(x)$ by $(x - b)$, and hence express $Q(x)$ as the product of a linear and a quadratic factor. **1**

(ii) Using your answer to part (i) or otherwise, simplify $\frac{(\tan^2 \theta - 1)(\sin \theta \cos \theta + 1)}{\cos^2 \theta (\tan^3 \theta - 1)}$. **2**

End of Question 4

For **Multiple Choice** questions, write down the correct option on your writing paper.

- (a) If $\cos \alpha = \frac{a}{b}$, where α is an acute angle, what is the value of $\sec(180^\circ + \alpha)$? 1

A. $\frac{\sqrt{b^2 - a^2}}{b}$

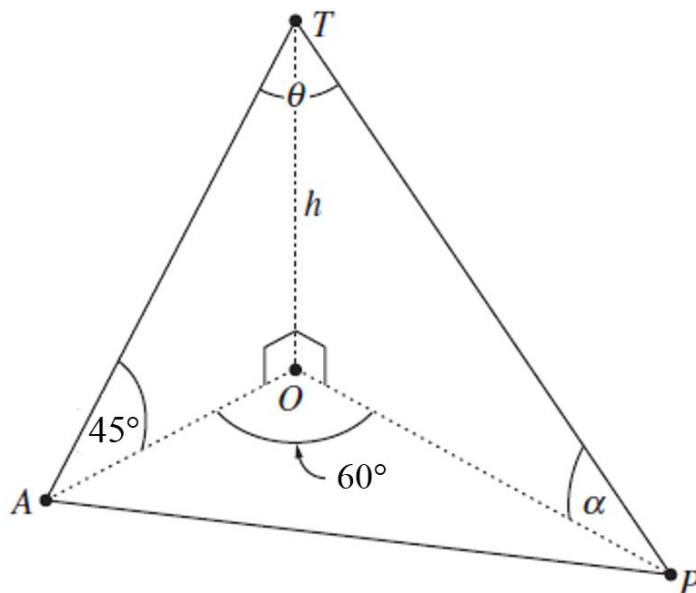
B. $\frac{a}{b}$

C. $-\frac{a}{b}$

D. $-\frac{b}{a}$

- (b) Find the domain of the function $f(x) = \sqrt{9 - x^2} + \frac{1}{x+3}$. 1

- (c) Consider the diagram below, which shows a vertical tower OT of height h metres, a fixed point A , and a variable point P that is constrained to move so that angle AOP is 60° . The angle of elevation of T from A is 45° . Let the angle of elevation of T from P be α and let angle ATP be θ .



- (i) By considering triangle AOP , show that $AP^2 = h^2 + h^2 \cot^2 \alpha - h^2 \cot \alpha$. 2
- (ii) By finding a second expression for AP^2 , deduce that $\cos \theta = \frac{1}{\sqrt{2}} \sin \alpha + \frac{1}{2\sqrt{2}} \cos \alpha$. 3

End of Assessment Task

Blank page



2021 Year 11 AT1 Mathematics Extension 1 - SOLUTIONS

Comments

Question 1

(a) (B) ✓

(b)(i) $\alpha + \beta = -\frac{b}{a}$
 $= -5$ ✓

(ii) $\alpha\beta = \frac{c}{a}$
 $= -1$ ✓

(iii) $(\alpha - \beta)^2 = \alpha^2 - 2\alpha\beta + \beta^2$
 $= \alpha^2 + \beta^2 - 2\alpha\beta$
 $= (\alpha + \beta)^2 - 4\alpha\beta$ ✓
 $= (-5)^2 - 4(-1)$
 $= 29$ ✓

(iv) $|\alpha - \beta| = \sqrt{(\alpha - \beta)^2}$
 $= \sqrt{29}$ ✓

Question 2

(a) (C) ✓

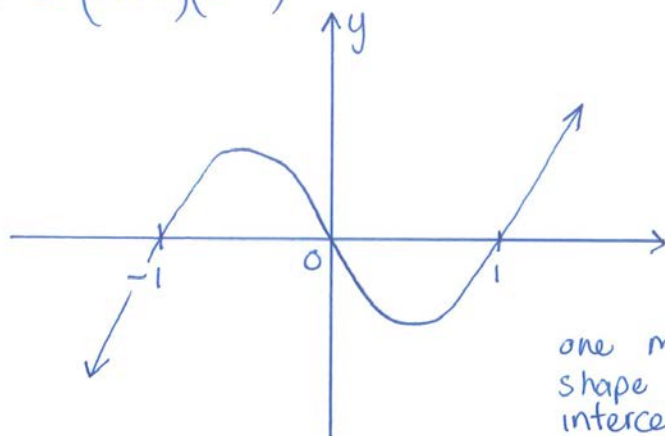
(b) $\angle L = 60^\circ$ $\frac{S}{\sqrt{T}} \frac{A}{C}$ ✓ one for correct placement
one for answers

$$-180^\circ \leq 2\theta \leq 180^\circ$$

$$2\theta = -120^\circ, -60^\circ$$

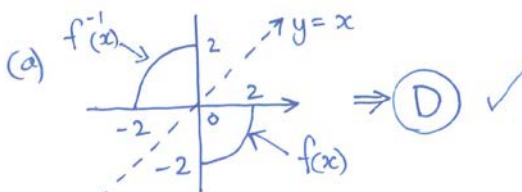
$$\therefore \theta = -60^\circ, -30^\circ$$

(c)(i) $y = x(x+1)(x-1)$



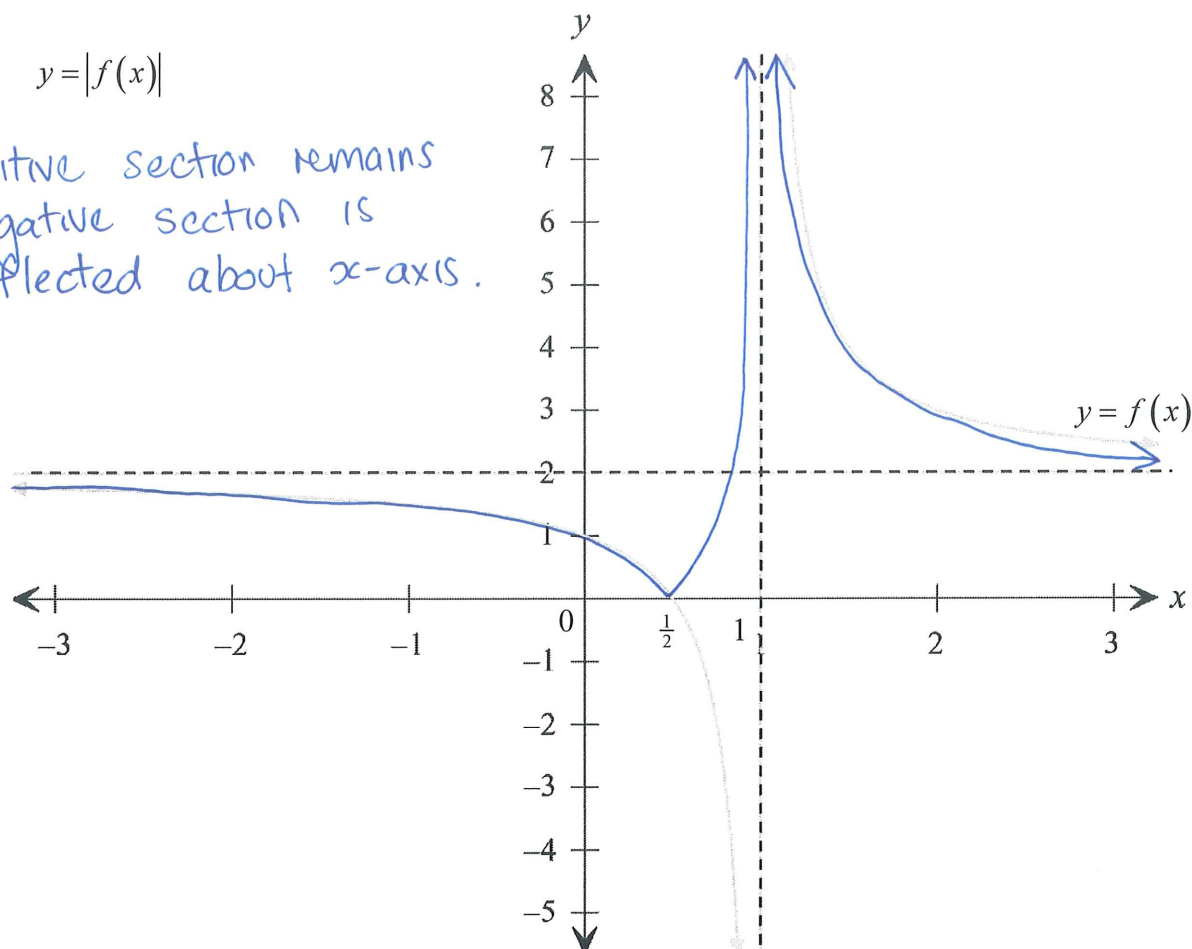
one mark for
shape and
intercepts.



Suggested Solution (s)	Comments
(ii) $\frac{x(x+1)(x-1)^2}{(x-1)} \geq 0$ multiplying by square of denominator $\Rightarrow x(x+1)(x-1) \geq 0$ using part (i) $-1 \leq x \leq 0$ and $x > 1$ one mark for each inequality. <u>Question 3</u> (a) $x^2 = \sin^2 \theta$ — (1) $y^2 = 4 \cos^2 \theta$ $\frac{y^2}{4} = \cos^2 \theta$ — (2) $(1) + (2) \Rightarrow x^2 + \frac{y^2}{4} = 1$ or $4x^2 + y^2 = 4 \Rightarrow (A)$ (b) separate solutions sheet	
<u>Question 4</u> (a)  $\Rightarrow (D) \checkmark$ (b) separate solutions sheet	

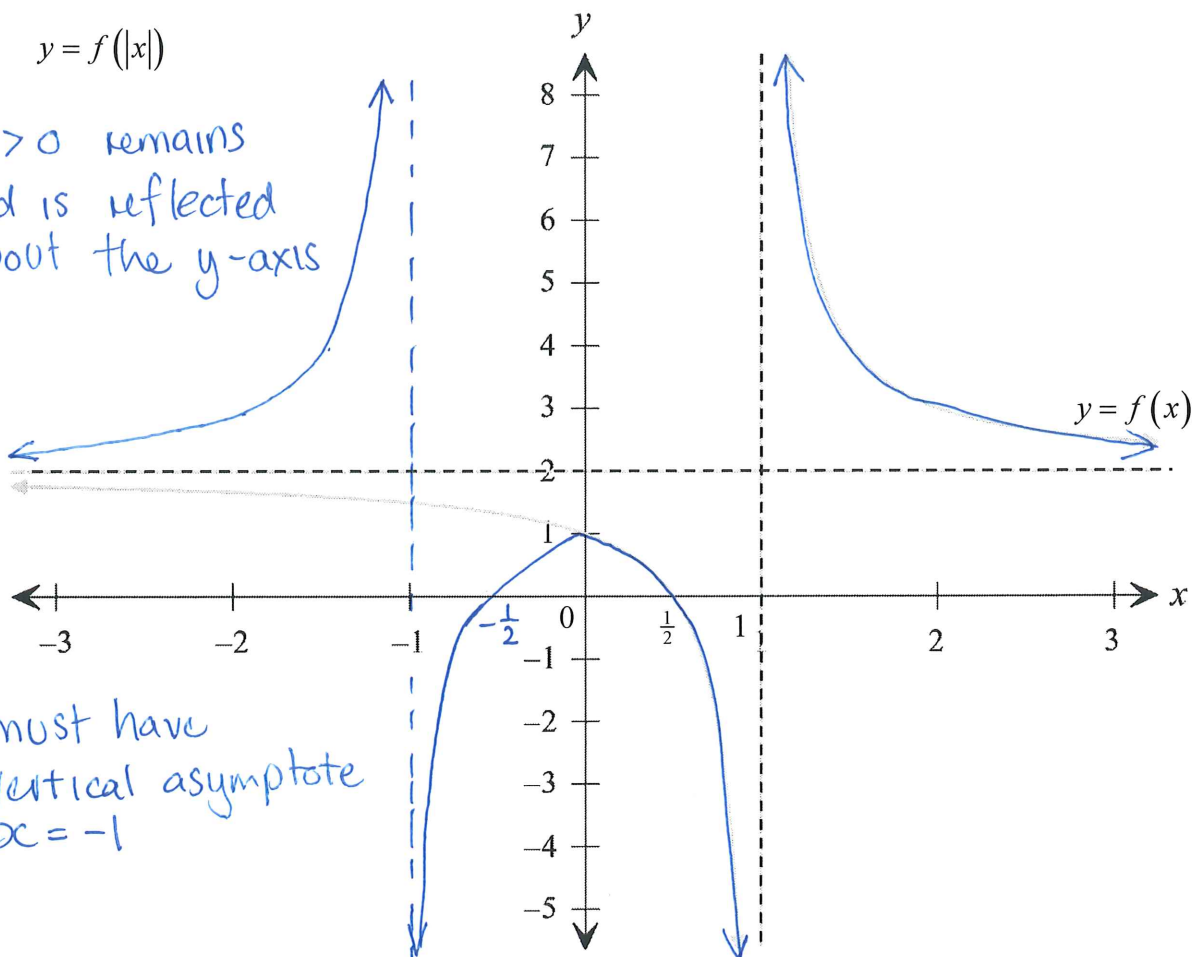
(b) (i) $y = |f(x)|$

- positive section remains
- negative section is reflected about x -axis.



(b) (ii) $y = f(|x|)$

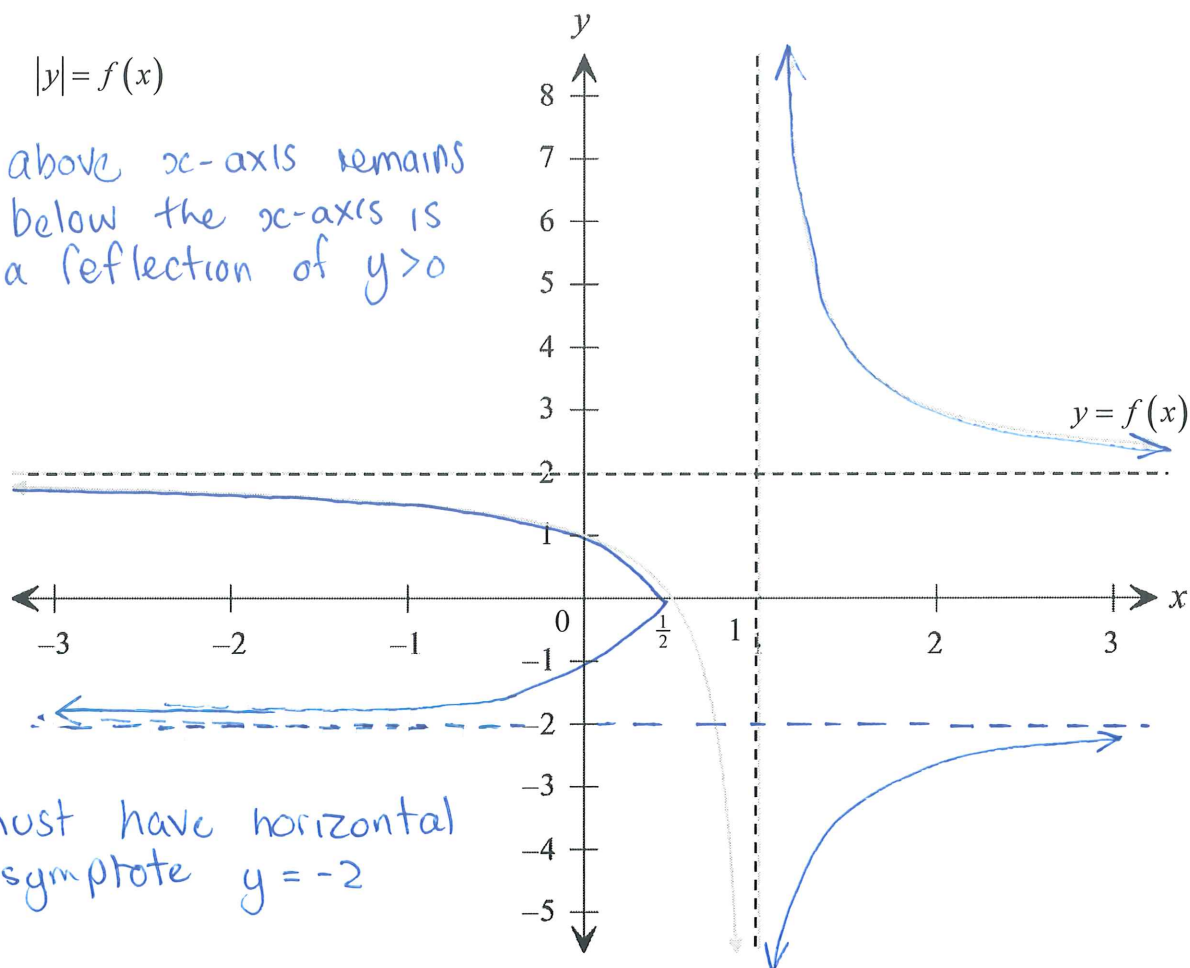
- $x > 0$ remains and is reflected about the y -axis



- must have vertical asymptote $x = -1$

(b) (iii) $|y| = f(x)$

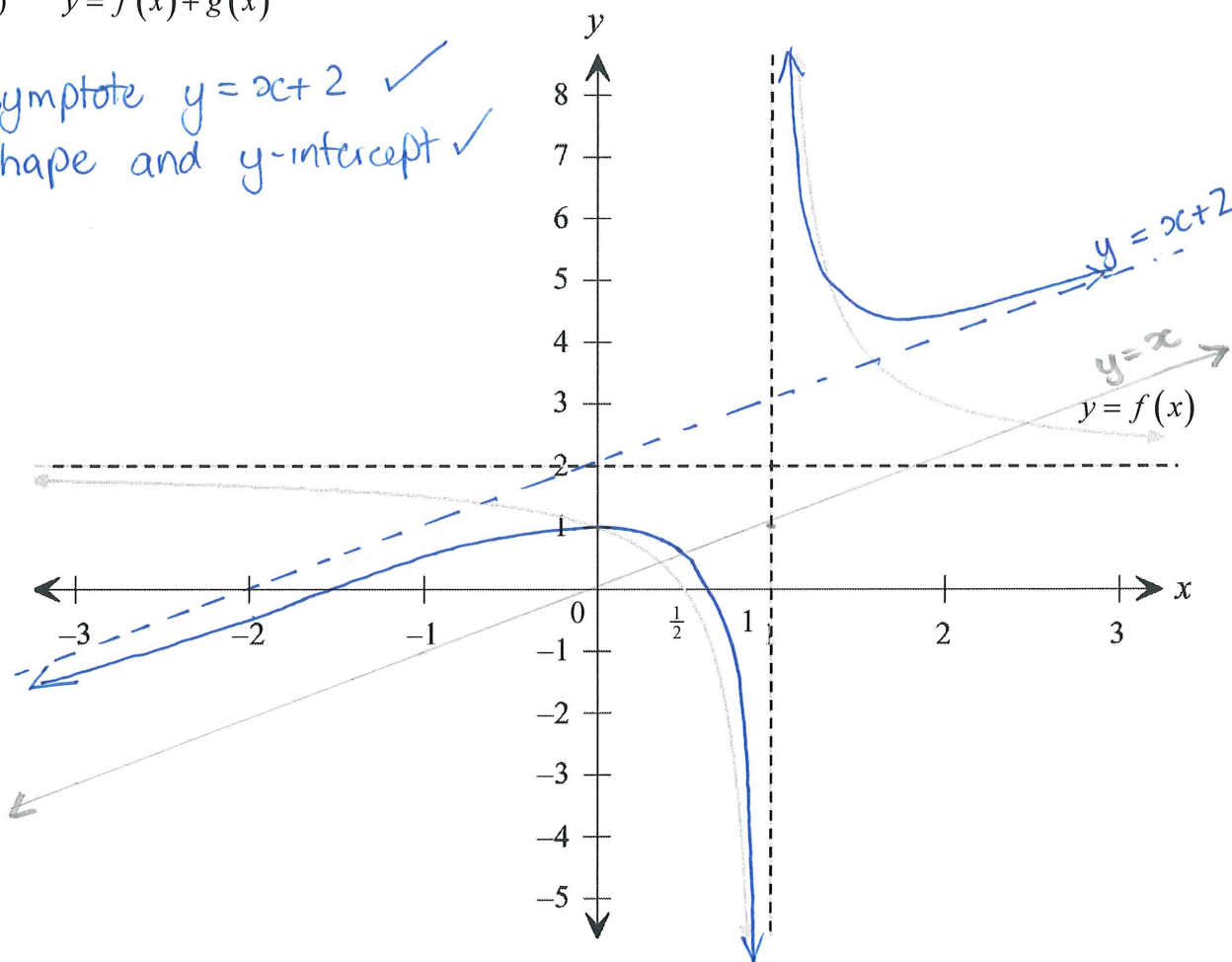
- above x -axis remains
- below the x -axis is a reflection of $y > 0$



- must have horizontal asymptote $y = -2$

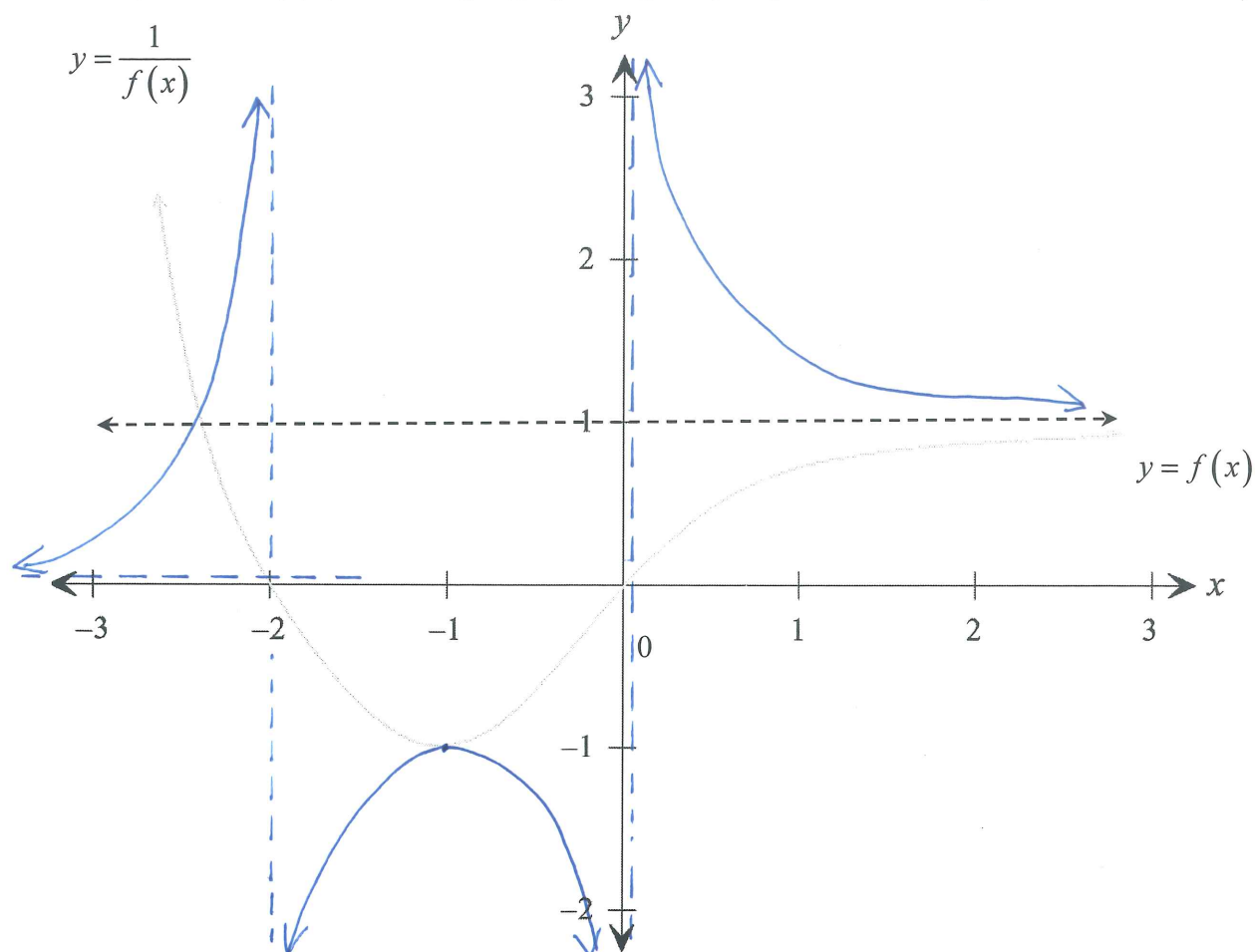
(b) (iv) $y = f(x) + g(x)$

- asymptote $y = x + 2$ ✓
- shape and y -intercept ✓

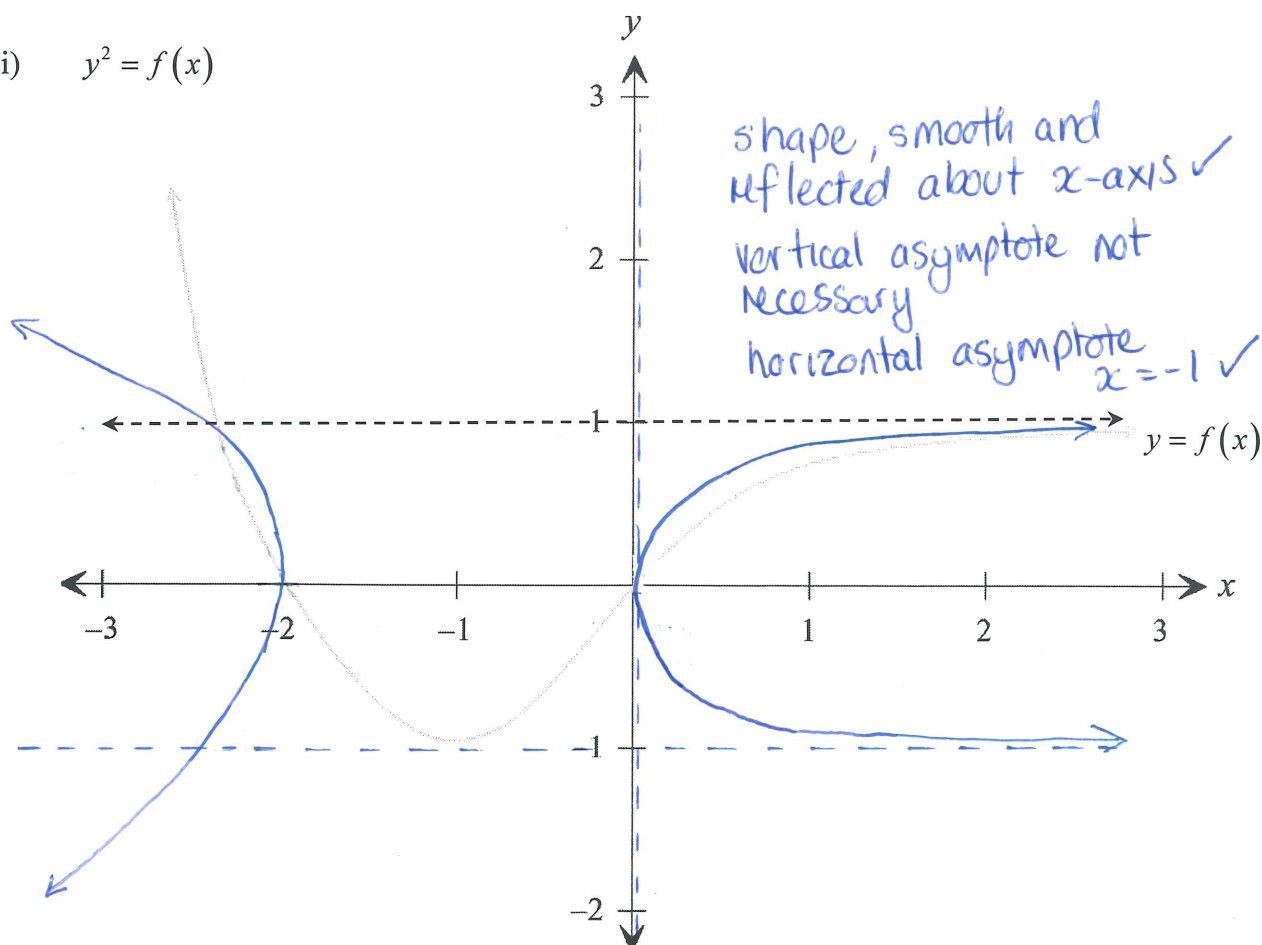


Answer Sheet for Question 4 (b) (Use the light grey as a guide, only lines made in pen will be marked).

(b) (i) $y = \frac{1}{f(x)}$

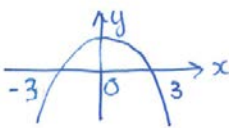


(b) (ii) $y^2 = f(x)$





2021 Year 11 AT1 Mathematics Extension 1 - SOLUTIONS

Suggested Solution (s)	Comments
(c)(i) $\begin{array}{r} x^2 + bx + b^2 \checkmark \\ x-b \overline{) x^3 + 0x^2 + 0x - b^3} \\ \underline{-(x^3 - bx^2)} \downarrow \\ bx^2 + 0x \\ \underline{-(bx^2 - b^3x)} \downarrow \\ b^3x - b^3 \\ \underline{-(b^3x - b^3)} \\ 0 \end{array}$ $\therefore x^3 - b^3 = (x-b)(x^2 + bx + b^2) \checkmark$ (ii) $\begin{aligned} \frac{(\tan^2 \theta - 1)(\sin \theta \cos \theta + 1)}{\cos^2 \theta (\tan^3 \theta - 1)} &= \frac{(\tan \theta + 1)(\tan \theta - 1)(\sin \theta \cos \theta + 1)}{\cos^2 \theta (\tan \theta - 1)(\tan^2 \theta + \tan \theta + 1)} \checkmark \\ &= \frac{(\tan \theta + 1)(\sin \theta \cos \theta + 1)}{\cos^2 \theta (\sec^2 \theta + \tan \theta)} \checkmark \\ &= \frac{(\tan \theta + 1)(\sin \theta \cos \theta + 1)}{1 + \sin \theta \cos \theta} \\ &= \tan \theta + 1 \end{aligned}$ <u>Question 5</u> (a) $\frac{s A}{\sqrt{T c}} \sec(180^\circ + \alpha)$ $\begin{aligned} &= \frac{1}{\cos(180^\circ + \alpha)} \\ &= \frac{-1}{\cos \alpha} \\ &= -\frac{b}{a} \Rightarrow \textcircled{D} \checkmark \end{aligned}$ (b) $9 - x^2 \geq 0$  $-3 \leq x \leq 3$ but $x \neq -3 \therefore D: -3 < x \leq 3 \checkmark$	



2021 Year 11 AT1 Mathematics Extension 1 - SOLUTIONS

Suggested Solution (s)	Comments
(c)(i) $\tan 45^\circ = \frac{h}{AO}$ $\tan \alpha = \frac{h}{OP}$ $AO = h$, $OP = h \cot \alpha$ both ✓ $AP^2 = AO^2 + OP^2 - 2(AO)(OP) \cos 60^\circ$ $AP^2 = h^2 + h^2 \cot^2 \alpha - 2h^2 \cot \alpha \left(\frac{1}{2}\right)$ } either ✓ (ii) $\sin 45^\circ = \frac{h}{AT}$, $\sin \alpha = \frac{h}{TP}$ $AT = \sqrt{2} h$, $TP = h \operatorname{cosec} \alpha$ $AP^2 = (\sqrt{2} h)^2 + (h \operatorname{cosec} \alpha)^2 - 2(\sqrt{2} h) h \operatorname{cosec} \alpha \cos \theta$ $= 2h^2 + h^2 \operatorname{cosec}^2 \alpha - 2\sqrt{2} h^2 \operatorname{cosec} \alpha \cos \theta$ ✓ $h^2 + h^2 \cot^2 \alpha - h^2 \cot \alpha = 2h^2 + h^2 \operatorname{cosec}^2 \alpha - 2\sqrt{2} h^2 \operatorname{cosec} \alpha \cos \theta$ dividing through by h^2 and using identity $1 + \cot^2 \alpha = \operatorname{cosec}^2 \alpha$ ✓ $1 + \cot^2 \alpha - \cot \alpha = 2 + \operatorname{cosec}^2 \alpha - 2\sqrt{2} \operatorname{cosec} \alpha \cos \theta$ $-\cot \alpha = 2 - 2\sqrt{2} \operatorname{cosec} \alpha \cos \theta$ $\frac{2\sqrt{2} \cos \theta}{\sin \alpha} = 2 + \frac{\cos \alpha}{\sin \alpha}$ } ✓ $2\sqrt{2} \cos \theta = 2 \sin \alpha + \cos \alpha$ $\cos \theta = \frac{1}{\sqrt{2}} \sin \alpha + \frac{1}{2\sqrt{2}} \cos \alpha$	